

The entire hex value is : 0xCD0A

The first byte is 0xCD @ XXXX

The second byte is 0x0A @ YYYY

Note:

XXXX & YYYY are some addresses.

Figure 9: Expected output of the code `bit_order.c` when run in big Endian architecture

2. Readability - Members can be easily addressed by the names assigned to them.
3. Low level programming – The biggest advantage of bit fields is that one does not have to keep track of how flags and masks actually map to the memory. Once the structure is defined, one is completely abstracted from the memory representation as in the case of bit-wise operations, during which one has to keep track of all the shifts and masks.

### Limitations

1. As we saw earlier, bit fields result in non-portable code. Also, the bit field length has a high dependency on word size.
2. Reading (using `scanf`) and using pointers on bit fields is

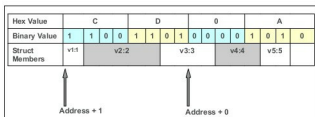


Figure 10: Bit ordering in big Endian architecture

- not possible due to non-addressability.
3. Bit fields are used to pack more variables into a smaller data space, but cause the compiler to generate additional code to manipulate these variables. This results in an increase in both space as well as time complexities.
4. The `sizeof()` operator cannot be applied to the bit fields, since `sizeof()` yields the result in bytes and not in bits. **END**

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